

ELMI CV

Elşad Allahyarov, DSc, PhD

Elmi işlər üzrə professor

Fizika departamenti, Keys Qərbi Rezerv Universiteti, Klivlend, Ohayo, Amerika Birləşmiş Ştatları

Əlaqə məlumatları və identifikatorlar

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Elm sahəsi	Nəzəri fizika; yumşaq maddə

Profil xülasəsi

Elşad Allahyarov Klivlenddəki Keys Qərbi Rezerv Universitetində fizika üzrə elmi işlər professorudur. Düsseldorfda Heinrich Heine Universitetində və Rusiya Elmlər Akademiyasının Yüksək Temperaturlar İnstitutunda da vəzifələri var. 2005-ci ildə Rusiya Elmlər Akademiyasında fizika və riyaziyyat üzrə elmlər doktoru dərəcəsi alıb. Dissertasiya The role of Coulomb correlations in colloids and biocolloids adlanır. CV bu dərəcəni alman Dr. habil. dərəcəsi ilə eyniləşdirir. 1994-cü ildə fizika üzrə fəlsəfə doktoru dərəcəsi alıb. CV bunu alman Dr. və ABŞ PhD dərəcəsi ilə eyniləşdirir. 1988-ci ildə Moskva Dövlət Universitetində fizika diplomu alıb. 2021–2025-ci illər üçün Alman Tədqiqat Cəmiyyətinin 300.000 avro luq qrantı kompozit elastomerlərin aktuasiyasına həsr olunub.

Təhsil

İllər	Dərəcə	Sahə / dissertasiya mövzusu	Təhsil müəssisəsi, şəhər, ölkə
2005	DSc	Fizika və riyaziyyat. Dissertasiya: The role of Coulomb correlations in colloids and biocolloids. CV bunu alman Dr. habil. dərəcəsi ilə eyniləşdirir.	Rusiya Elmlər Akademiyası, Moskva, Rusiya
1994	PhD	Fizika. Dissertasiya: Structure factor and effective interactions in non-ideal plasma and liquid metals. CV bunu alman Dr. və ABŞ PhD dərəcəsi ilə eyniləşdirir.	Rusiya Elmlər Akademiyası, Moskva, Rusiya
1988	Fizika diplomu	Dissertasiya: Injection of relativistic electron beams into dense plasma. CV bunu fizika üzrə Master of Arts ekvivalenti adlandırır.	Moskva Dövlət Universiteti, Moskva, Rusiya

Elmi və peşəkar vəzifələr

İllər	Vəzifə	Kafedra / bölmə	Təşkilat, şəhər, ölkə
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İllər	Vəzifə	Kafedra / bölmə	Təşkilat, şəhər, ölkə
	Elmi işlər üzrə professor		Yumşaq Maddə İnstitutu, Nəzəri Fizika II, Heinrich Heine Universiteti, Düsseldorf, Almaniya
	Professor		Nəzəri şöbə, Yüksək Temperaturlar İnstitutu, Rusiya Elmlər Akademiyası, Moskva, Rusiya
2018–2025	Elmi işlər üzrə professor		Fizika departamenti, Keys Qərbi Rezerv Universiteti
2012–2017	Tədqiqatçı		Nəzəri Fizika II İnstitutu, Heinrich Heine Universiteti, Düsseldorf
2014 and 2015	Hər il altı ay qonaq professor		Makromolekulyar elm, Keys Qərbi Rezerv Universiteti
2006–2011	Elmi işçi		Fizika departamenti, Keys Qərbi Rezerv Universiteti
2004–2006	Postdoktorant		Nəzəri Fizika II İnstitutu, Heinrich Heine Universiteti, Düsseldorf
2000–2003	Privatdozent		Bərk Cisim Tədqiqatı İnstitutu, IFF Theory II, Yulix Nüvə Tədqiqat Mərkəzi, Almaniya
1995–1997	Qonaq tədqiqatçı		Mürəkkəb mayelər laboratoriyası, Eyndhoven Texnologiya Universiteti, Niderland
1988–1999	Aspirant və tədqiqatçı		Yüksək Temperaturlar İnstitutu, Rusiya Elmlər Akademiyası, Moskva

Nəşrlər

Rəylənmiş (peer-reviewed) jurnal məqalələri

1. Dispersion of plasma oscillations in non-ideal Coulomb systems **E. Allahyarov** , S. A. Trigger, Teplofizika Vysokikh Temperatur 30, 136 (1992).
2. Three-dimensional dispersion of Langmuir oscillations in non-ideal plasma **E. Allahyarov** , S. A. Trigger, Teplofizika Vysokikh Temperatur 30, 462 (1992).
3. The influence of non-Coulomb interaction on plasma oscillation spectrum in classical two-component plasma **E. Allahyarov** , V. B. Bobrov, S. A. Trigger, Teplofizika Vysokikh Temperatur 30, 876 (1992).
4. On the temperature dependence of the liquid lithium electric resistance, **E. Allahyarov** , V. B. Bobrov, Teplofizika Vysokikh Temperatur 31, 207 (1993)
5. Static structure factor and electric conduction of expanded liquid cesium, **E. Allahyarov** , V. B. Bobrov, S. A. Trigger, Teplofizika Vysokikh Temperatur 31, 44 (1993).
6. Electrical conductivity of fully ionized plasmas with arbitrary electron degeneracy. 1: Classical plasma, V. B. Bobrov, **E. Allahyarov** , High temperatures 31, 422 (1993).
7. Electric conductivity of a fully ionized plasma with arbitrary electron degeneration. 2. Liquid-metal plasma V. B. Bobrov, **E. Allahyarov** , Teplofizika Vysokikh Temperatur 31, 352 (1993).
8. Kramers–Kronig relations for the dielectric constant of a classical plasma and the spectrum of plasma oscillations. **E. Allahyarov** , V.B. Bobrov, High Temperatures 31, 644 (1993)
9. Structure factor of liquid metals at small wave vectors and a model of a one-component plasma, **E. Allahyarov** , V. B. Bobrov, S. A. Trigger, High Temperatures 32, 341 (1994).
10. OCP reference theory of dynamic structure factors of expanded liquid metals S. A. Trigger, **E. Allahyarov** , P. P. J. M. Schram, Physica B 212, 97 (1995).

11. Low-angle structure behavior of expanded liquid metals, E. Allahyarov , P. P. J. M.Schram, S. A. Trigger, *Physica B* 212, 88 (1995).
12. Overscreening in colloidal systems: the attractive branch of pair macroion interaction, **E. Allahyarov** , L. I. Podloubny, P. P. J. M.Schram, S. A. Trigger, *Physica A* 220, 349 (1995).
13. MD simulation of the pair correlation function and static structure of charged colloidal suspensions, **E. Allahyarov** , P. P. J. M. Schram, S. A. Trigger, *Phys.B*, 228, 136 (1996).
14. Effective macroion interaction in colloidal systems: Possibility of attraction, **E. Allahyarov** , L. I. Podloubny, P. P. J. M. Schram, S. A. Trigger, *Physica B* 229, 132 (1996).
15. Damping of longitudinal waves in charged colloids, **E. Allahyarov** , L. I. Podloubny, P. P. J. M. Schram, S. A. Trigger, *Physica B* 228, 166 (1996).
16. The breaking crack build-up in perforated planes by uniform ring switching, V. M. Mirsalimov, **E. Allahyarov** , *Int. J. Fracture* 79, R17 (1996).
17. Damping of longitudinal waves in colloidal crystals of finite size **E. Allahyarov** , L. I. Podloubny, P. P. J. M.Schram, S. A. Trigger, *Phys. Rev. E* 55, 592 (1997).
18. Attraction between like-charged macroions by Coulomb depletion **E. Allahyarov** , I. D'Amico, H. Löwen, *Phys. Rev. Letters* 81, 1334 (1998).
19. Possibility of an attraction between highly charged colloids **E. Allahyarov** , P. P. J. M. Schram, S. A. Trigger, *Proc. Int. Conf. Strongly Coupled Coulomb Systems*, Plenum Press, New-York, 247 (1998).
20. Role of effective triplet interactions in charged colloidal suspensions H.Löwen, **E. Allahyarov** , *J. Phys. Condensed Matter* 10, 4147 (1998).
21. Effective forces between macroions: the cases of asymmetric macroions and added salt **E. Allahyarov** , H. Löwen, S. Trigger, *Phys. Rev. E* 57, 5818 (1998).
22. Effect of confinement on the interaction between charged colloidal particles **E. Allahyarov**, I. D'Amico, H. Löwen, *Phys. Rev. E* 60, 3199 (1999).
23. The hard physics of soft matter H. Löwen, **E. Allahyarov**, et al, *Advances in Solid State Physics* 40, 809 (2000).
24. Effective interaction between helical biomolecules **E. Allahyarov** , H. Löwen, *Phys. Rev. E* 62, 5542 (2000).
25. Effective interaction between confined colloids: repulsion or attraction H. Löwen, **E. Allahyarov** , I. D'Amico, *Progr. Colloid Polym. Sci.* 115, 367 (2000).
26. Influence of solvent granularity on the effective interaction between charged colloidal suspensions **E. Allahyarov** , H. Löwen, *Phys. Rev. E* 63, 041403 (2001).
27. Discrete solvent effects on the effective interaction between charged colloids **E. Allahyarov**, H. Löwen, *J. Phys. Condensed Matter* 13, L277 (2001).
28. Interactions and phase transitions of colloidal dispersions in bulk and at interfaces H. Löwen, **E. Allahyarov** , et al, *Phil. Trans. Roy. Soc. Lond. A* 359, 909 (2001).
29. Discrete charge patterns, Coulomb correlations and interactions in protein solutions **E. Allahyarov** , H. Löwen, J. P. Hansen, A. A. Louis, *Europhys. Letters*. 57, 731 (2002).
30. Effective forces in colloidal mixtures: from depletion attraction to accumulation repulsion A. A. Louis, **E. Allahyarov** , H. Löwen, R. Roth, *Phys. Rev. E* 65, 061407 (2002).
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33. Charged colloids, polyelectrolytes and biomolecules viewed as strongly coupled Coulomb systems H. Löwen, **E. Allahyarov** , et al., *J. Phys. A: Math. Gen.* 36, 5827 (2003).
34. Mesoscopic solvent simulations: Multi-particle-collision dynamics of three-dimensional flows **E. Allahyarov** , G. Gompper, *Phys. Rev. E* 66, 036702 (2003).
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36. Re entrance in DNA condensation, **E. Allahyarov** , N. Gusein-zade, *Phys. Reports of Russian Academy of Sciences* 6, 49 (2004).

37. Crystallization of colloidal plasma: charge renormalization in added salt **E. Allahyarov** , S. Trigger, High Temperature Physics 43, 315 (2005).
38. Attraction between DNA molecules mediated by multivalent ions **E. Allahyarov** , G. Gompper, H. Löwen, Phys. Rev. E. 69, 041904 (2004); Virtual Journal of Biological Physics Research 7, (2004).
39. DNA condensation, re dissolution and mesocrystals induced by tetravalent counterions **E. Allahyarov** , H. Löwen, G. Gompper, Euro. Phys. Letters 68, 894 (2004).
40. Interaction between overcharged DNA molecules **E. Allahyarov** , G. Gompper, H. Löwen, J. Phys. Cond. Matter 17, S1827 (2005).
41. Charged colloids and polyelectrolytes: from statics to electrokinetics H. Löwen, **E. Allahyarov** et al, J. Phys.: Conf. Ser. 11, 207 (2005).
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44. Charged colloidal particles in a charged wedge: do they go in or out H. Löwen, **E. Allahyarov** et al, J. Phys.: Condens. Matter 20, 404221 (2008).
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49. Simulation Study of Sulfonate Cluster Swelling in Ionomers **E. Allahyarov** , P. Taylor, H. Löwen, Phys. Rev. E 80, 061802 (2009).
50. Simulation study of field-induced morphological changes in a proton-conducting ionomer **E. Allahyarov** , P. Taylor, H. Löwen, Phys. Rev. E 81, 031805 (2010).
51. Simulation study of the equilibrium morphology in ionomers with different architectures **E. Allahyarov** , P. Taylor, J. Polymer Sci. B: Polymer Physics 49, 368 (2011).
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53. Heterogeneous crystallization of hard-sphere colloids near a wall K. Sandomirski, **E. Allahyarov** , H. Löwen, S. U. Egelhaaf, Soft Matter 7, 8050 (2011).
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60. Highly asymmetric electrolytes in the primitive model: Hypernetted chain solution in arbitrary spatial dimensions. M. Heinen, **E. Allahyarov** , H. Löwen, J. Comp. Chem. 35, 275-289 (2014).
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62. Heterogeneous crystallization of hard and soft spheres near flat and curved walls, K. Sandomirski, S. Walta, J. Dubbert, **E. Allahyarov**, A. B. Schofield, H. Löwen, W. Richtering, S. U. Egelhaaf, *Eur. Phys. J* 223, 439-454 (2014).
63. Analysis of the actuation properties of charged multilayer films, **E. Allahyarov**, H. Löwen and L. Zhu, *J. Appl. Phys.* 117, 034504 (1-14) (2015).
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65. A simulation study of the electrostriction effects in dielectric elastomer composites containing polarizable inclusions with different spatial distributions, **E. Allahyarov**, H. Löwen, L. Zhu, *Phys. Chem. Chem. Phys.* 17, 32479-32497 (2015).
66. Semicrystalline structure - dielectric property relationship and electrical conduction in a biaxially oriented poly(vinylidene fluoride) film under high electric fields and high temperatures, Yang, J. Ho, **E. Allahyarov**, R. Mu, L. Zhu, *ACS Appl. Mater. Interfaces* 7, 19894-19905 (2015).
67. Enhanced ionic diffusion in ionomer-filled nanopores, **E. Allahyarov**, P. L. Taylor, and H. Löwen, *J. Chem. Phys.* 143, 243126 (1-10) (2015).
68. Interfacial Polarization-Induced Loss Mechanisms in Polypropylene/BaTiO₃ Nanocomposite Dielectrics, G. Zhang, D. Brannum, D. Dong, L. Tang, **E. Allahyarov**, S. Tang, Saide, K. Kodweis, J.-K. Lee, L. Zhu, *Chemistry of Materials* 28, 4646-4660 (2016).
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70. Simulation Study of Ion Diffusion in Charged Nanopores with Anchored Terminal Groups, **E. Allahyarov**, H. Löwen, P. L. Taylor, *Electrochimica Acta* 242, 73-85 (2017).
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73. Understanding reversible Maxwellian electroactuation in a 3M VHB dielectric elastomer with prestrain, L. Liu, Y. Huang, Y. Zhang, **E. Allahyarov**, Z. Zhang, F. Lv, L. Zhu, *Polymer* 144, 150-158 (2018).
74. Polymer Nanodielectrics: Current Accomplishments and Future Challenges for Electric Energy Storage, G. Zhang, **E. Allahyarov**, L. Zhu, Springer Nature, Editors B. Li and T. Jiao, 1-48, (2018)
75. Length segregation in mixtures of spherocylinders induced by imposed topological defects, **E. Allahyarov**, H. Löwen, *Soft Matter* 14, 8962-8973 (2018)
76. Reducing dielectric loss and enhancing electrical insulation for multilayer polymer films by nanoconfined ion transport under high poling electric fields, **E. Allahyarov** et al, *J. Mater. Chem. C* 1-20 (2020), <https://doi.org/10.1039/D0TC00522C>
77. Reducing Dielectric Loss by Nanoconfined Impurity Ion Transport in Multilayer Films under Low Electric Fields, **E. Allahyarov** et al, *Composites Part B- Engineering* 190, 107908 1-16 (2020).
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85. **Elshad Allahyarov**, Hartmut Löwen, Alan R. Denton, Structural correlations in highly asymmetric binary charged colloidal mixtures, Phys. Chem. Chem. Phys. 25, 15439-15451 (2022).
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93. **Elshad Allahyarov** and Hartmut Lowen, Electric field-induced clustering in nanocomposite films of highly polarizable inclusions, J. Colloid & Interface Science 668, 587-598 (2024).
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Qrantlar və tədqiqat layihələri

İllər	Layihənin adı	Maliyyələşdirən qurum	Rolunuz
2021–2025	Kompozit elastomerlərin aktuasiya xassələri. 300.000 avro	Alman Tədqiqat Cəmiyyəti (DFG)	

Tədris fəaliyyəti

İllər	Kurs	Səviyyə	Təhsil müəssisəsi
1990–1998	Nəzəri mexanika, optika, elektromaqnetizm, kvant mexanikası, statistik mexanika, mayelər nəzəriyyəsi, termodinamika, bərk cisim fizikası, plazma fizikası, hidrodinamika və riyaziyyat		Moskva Energetika İnstitutu
2000–2004	Yumşaq maddə fizikası, statistik fizika və termodinamika		Düsseldorf Universiteti, Almaniya
2009–2010	PHYS 116, Introduction to Physics II		Case Western Reserve University

İllər	Kurs	Səviyyə	Təhsil müəssisəsi
2017–2020	Kvant molekulyar dinamika, CP2K, Kar–Parrinello və DFT modelləşdirməsi		Duisburq-Essen Universiteti, Almaniya